

CAD Office Workspace Design

Student	Naman Arora - Electrical Engineering
Course context	Engineering 1030 - Engineering Graphics / first-year CAD learning project
Project type	Office workspace CAD model, floor plan, sections, elevations, perspectives, drawing sheets
Portfolio positioning	CAD fluency, engineering graphics, spatial reasoning, design communication, foundation for electrical design

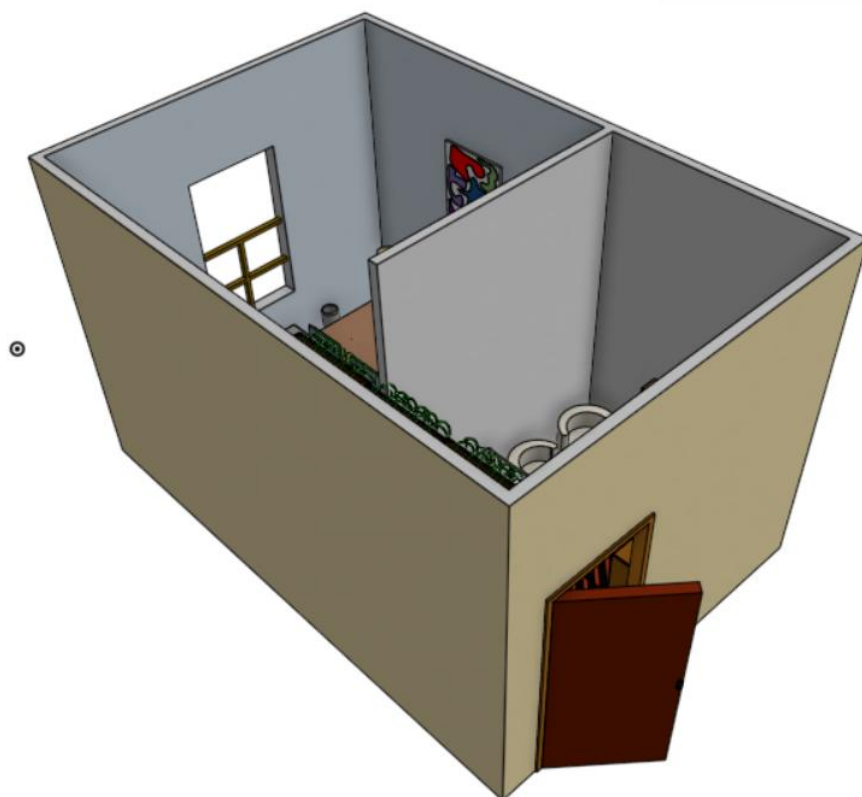


Figure 1. 3D cutaway perspective of the office concept, showing the entry, partition wall, green-wall feature, work zone, and rest zone.

Honest portfolio angle

This is presented as an engineering graphics and CAD documentation project, not as a code-compliant architectural or electrical installation package. Its strength is that it shows early design process, drawing literacy, and communication discipline that support later electrical design work.

1. Research-backed portfolio positioning

How this project can help you market yourself online without overstating it.

Engineering portfolios are useful because they let candidates go deeper than a resume and can include CAD drawings, photos, simulations, test graphics, and other work samples that demonstrate skills in action.

A website portfolio should be built around a clear goal and audience. For your target direction - electrical design, E&I, controls, automation, and engineering co-op roles - this project should quickly show CAD fluency and communication quality before going into details.

A strong project case study should present a realistic design challenge, appropriate complexity, the design process, and the reasoning behind the solution rather than only showing final images.

This is why the website version should use a compact case-study flow: problem, constraints, solution, implementation, results, skills demonstrated, and what you would improve next.

How it helps your engineering brand

- Shows that your technical foundation started early, before later power systems, industrial, data, automation, and PCB projects.
- Adds a visual project to your portfolio, which makes the site easier for recruiters to scan and remember.
- Demonstrates technical communication: you created plans, sections, elevations, 3D views, and presentation sheets instead of only a render.
- Connects to electrical engineering design work through spatial layout, drawing interpretation, equipment placement, documentation discipline, and coordination with building constraints.

2. Problem definition

What the project was trying to solve.

The project challenge was to create a functional office workspace inside a compact room envelope and communicate the design using CAD-generated views. The model needed to include the physical room, internal partition, door, window, desk layout, seating, storage, rest area, visual elements, and presentation drawings.

Design problem statement

Problem

Design a compact office environment that balances focused work, storage, rest seating, circulation, daylight, and visual identity, then document it clearly enough that the layout can be understood from plan, section, elevation, and 3D perspective views.

Key constraints interpreted from the model

- Fixed rectangular room boundary with existing exterior walls.
- Entry door and window locations create circulation and daylight constraints.
- Internal partition separates the work/rest side from the support/storage side.
- Furniture must fit without blocking the entry, work area, or rest area.
- The final output must be presentation-ready, not just a raw CAD model.
- Scale and drawing-sheet discipline were required in the course deliverables, including section drawings at 1:50 scale.

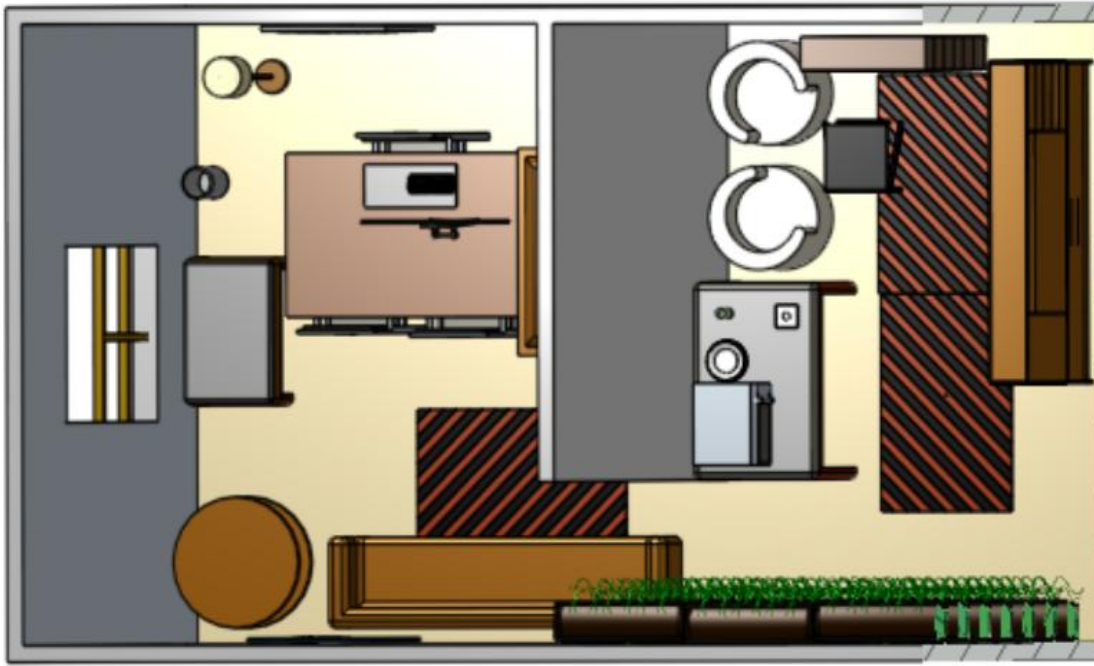


Figure 2. Floor plan showing the overall zoning: work desk, seating, storage, rest area, green-wall feature, window, and door relationship.

3. Design solution

How the final office layout solves the problem.

The final layout uses zoning to make the room understandable. The main desk area is placed as the primary productivity zone, while a support/rest zone provides seating, storage, and secondary utility surfaces. A green-wall feature adds visual interest and helps organize the left side of the space.

Work zone	Support/rest zone	Green-wall/storage zone
Desk, monitor, chairs, shelves, lamps, wall art, and daylight through the window create a focused workspace.	Lounge chairs, storage, and secondary surfaces support breaks, informal review, and additional storage.	The plant wall visually softens the technical space while separating storage and utility elements from the work zone.

Why this solution is effective

- The office is not a random furniture layout; it has readable functional zones.
- The design uses the window and wall surfaces to create a more realistic workspace environment.
- The internal partition gives the model a clear architectural constraint, making the CAD problem more complex than a single open room.
- The project includes both presentation views and technical views, which is important for communicating engineering work to different audiences.

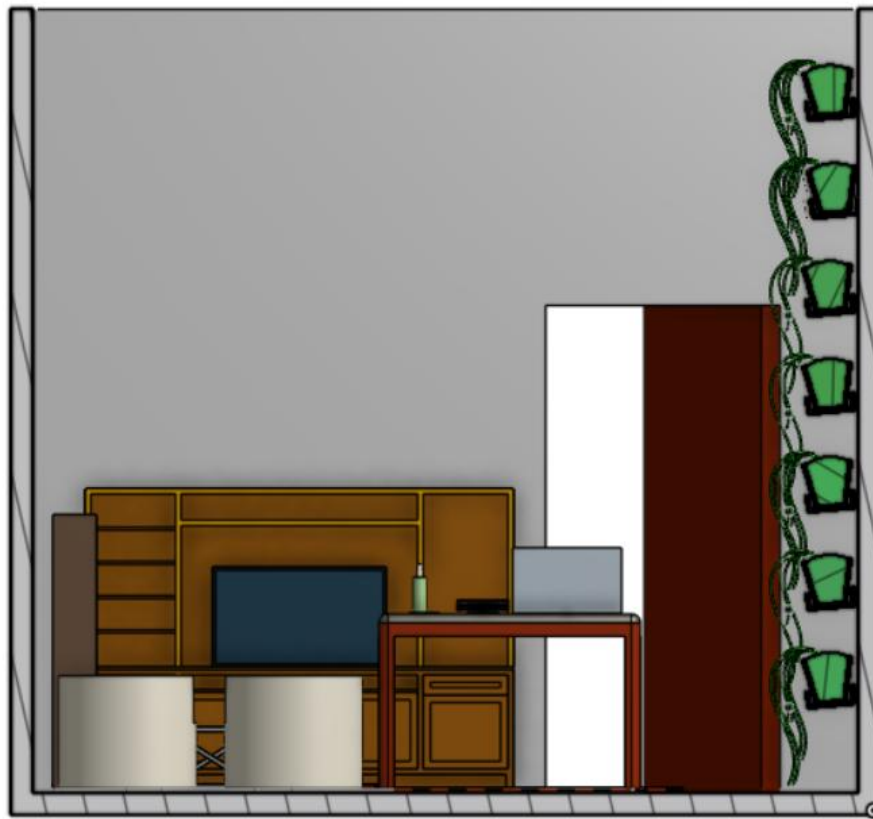


Figure 3. Rest/support side of the workspace, showing the green-wall feature, lounge seating, storage, and utility surfaces.

4. Implementation process

The work sequence behind the CAD model and drawing package.

A useful engineering portfolio document should show the process, not only the final render. The engineering design process typically starts by identifying the exact problem, objectives, motivations, and deliverables, then develops specifications and moves through conceptual and detailed design before delivery documentation.

1. Created the room envelope: modelled the exterior walls, floor boundary, open top, and visible wall thickness.
2. Added openings: placed the entry door and window, then checked how these openings affected circulation and furniture placement.
3. Built the internal partition: divided the room into a main office/work zone and a support/rest/storage zone.
4. Inserted furniture and objects: desk, chairs, cabinet, storage units, shelves, lamps, rugs, tables, lounge seating, plant wall, and artwork.
5. Applied materials and visual hierarchy: used different wall colors, furniture tones, rugs, plants, and accent elements to make the design readable.
6. Generated documentation views: floor plan, section views, elevations, cutaway perspectives, and drawing sheets with title-block style presentation.
7. Reviewed consistency across views: checked whether the same layout was understandable from plan, section, and 3D perspectives.

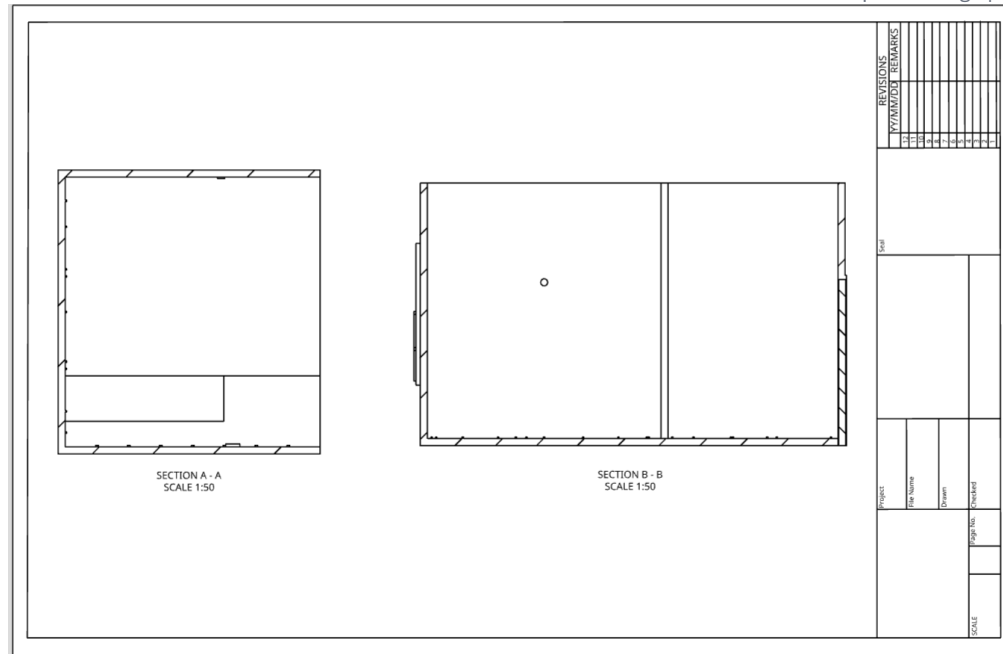


Figure 4. Course drawing sheet showing section views A-A and B-B at 1:50 scale.

5. Deep analysis: value for an electrical engineering student

What this project says about your technical development.

For an electrical engineering portfolio, the most important framing is that this project is a bridge between visual design and technical design. It does not prove electrical design calculations by itself, but it does prove early comfort with CAD, drawing views, spatial coordination, and technical documentation.

Electrical design involves planning and creating components, schematics, lighting equipment, power systems, and telecommunications infrastructure; CAD tools are used to create electrical drawings, schematics, wiring diagrams, and other design documentation.

The connection is straightforward: before an engineer can place devices, route cable, create lighting zones, or coordinate equipment, they must understand the physical space and communicate it clearly through drawings.

Transferable engineering skills demonstrated

CAD / graphics skill	Electrical engineering relevance
Plan, elevation, and section views	Useful for reading building layouts, panel locations, cable routes, lighting layouts, and equipment clearances.
3D spatial reasoning	Supports coordination between electrical devices, furniture/equipment, walls, doors, windows, and maintenance access.
Drawing-sheet discipline	Relates to professional deliverables such as single-line drawings, schematics, panel layouts, and field markups.
Material and object organization	Builds the habit of separating systems, layers, components, and design intent for easier review.
Design communication	Shows ability to explain technical work to recruiters, instructors, non-technical stakeholders, and engineering reviewers.

ABET-aligned interpretation

ABET describes engineering design as an iterative process of devising a system, component, or process to meet desired needs and specifications within constraints. It also emphasizes student outcomes such as solving engineering problems, applying engineering design, and communicating effectively with different audiences.

This project aligns with that mindset at an introductory level: you identified a design need, worked within spatial constraints, selected a layout, created a detailed model, and communicated the result through multiple drawings.



Figure 5. Back-wall section view. This view demonstrates cross-sectional communication, plant-wall organization, storage placement, and work-zone relationship.

6. Future electrical design extension

How to make this project stronger for electrical design roles.

The strongest way to upgrade this project for an electrical engineering website would be to add a second layer of design documentation that treats the office as a small electrical space-planning problem. This would make the project more directly relevant to electrical design, building systems, and controls roles while preserving the original first-year CAD story.

Extension	What to add	Why it matters
Lighting plan	Ceiling fixtures, task lighting, switch zones, daylight consideration.	Connects CAD layout to lighting design and user comfort.
Power layout	Duplex receptacles, workstation power, printer/utility power, dedicated loads if needed.	Shows practical electrical planning around furniture and equipment.
Low-voltage/data	Ethernet/data outlet, Wi-Fi/AP concept, cable path, desk connectivity.	Links to controls, IT, automation, and workplace systems.
Load estimate	Small load table for computer, printer, lighting, and miscellaneous devices.	Adds engineering calculation discipline without overcomplicating the project.
Controls/automation	Occupancy sensor, daylight dimming idea, smart switch, energy-saving concept.	Connects to automation, sensors, and energy efficiency.
Drawing set update	Separate sheet for architectural plan and electrical annotations.	Makes the deliverable more professional and easier to review.



Figure 6. Window-side section view, useful for discussing future lighting, task-lighting, receptacle, data, and workstation coordination.

7. Outcomes and skills demonstrated

What the project produced and what it proves.

Project outcomes

- Produced a complete office CAD model with walls, partition, window, door, furniture, decorative elements, and clear zones.
- Generated a top floor plan for spatial organization and circulation review.
- Generated section/elevation views to show vertical relationships, furniture placement, and wall features.
- Created cutaway perspective views and drawing sheets that make the project easy to understand visually.
- Practiced presentation drawing discipline through title-block sheets and scale-based section views.

Skills to list on the website

CAD modelling	Engineering graphics	Technical drawing	Spatial design
Section views	Presentation sheets	Design documentation	Portfolio storytelling

